

7.6 Fenchel-Nielsen Coordinates on Teichmüller Space

FN coordinate: length + twist

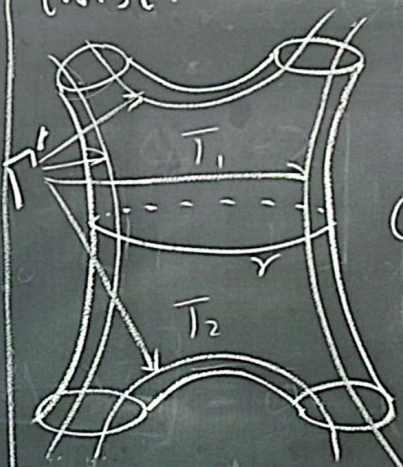
length: S is cpt finite type, choose a trouser decomposition by maximal multicurve $\Gamma = \{\gamma_1, \dots, \gamma_{3g-3}\}$

$\forall \tau \in T$ on S , $\varphi: S \rightarrow X$ representing a point $\tau \in T_S$

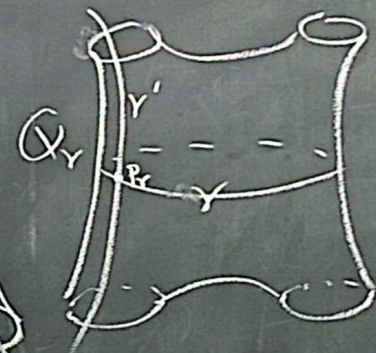
$\exists!$ geodesic on X in the homotopy class of $\varphi(\gamma)$ on X

Denote the length of this geodesic by $l_\gamma(\tau)$, define a map $FN_L: T_S \rightarrow (\mathbb{R}_+^*)^\Gamma$ by $FN_L(\tau) = (l_\gamma(\tau))_{\gamma \in \Gamma}$

twist:



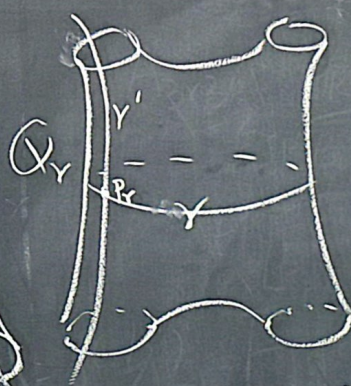
let Q_γ be the universal cover of S with γ as one boundary component



$\varphi(\gamma), \gamma \in \Gamma$,
hyperbolic tra
the trousers
 $[\varphi(Q_\gamma)], [\varphi(Q_\gamma)]$

twist:

let Q_r be the union of (one or two) trousers with r as one boundary component.



$r' \subset Q_r$ of one of the curves in Γ' joining one point of the boundary of Q_r to another, intersects r once at p_r

$\varphi(r)$, $r \in \Gamma'$, cut $X = \varphi(S)$ into hyperbolic trousers, corresponding to the trousers of S . We get $[\varphi(\Gamma)]$, $[\varphi(Q_r)]$, $[\varphi(r)]$

$\exists ! \alpha_r$ homotopic to $[\varphi(r)]$, joining boundary points of $[\varphi(Q_r)]$, and consist:

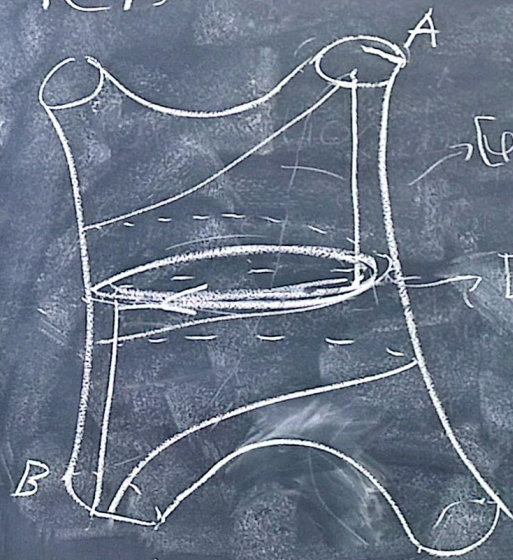
1. minimal geodesic joins $[\varphi(r)]$ to the component of $\partial[\varphi(Q_r)]$ containing one of ends of $\varphi(r')$
2. an immersed geodesic S_r , whose image $C[\varphi(r)]$, length = $t_r(\tau)$
3. minimal geodesic α_r joining the other end of $\varphi(r')$



$$\text{length } S_r = t_r(\tau)$$

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Def: The twist coordinate map: $FN_T: T_S \rightarrow \mathbb{R}^g$ associate to $T \in T_S$ the vector $(t_\gamma(T))_{\gamma \in \Gamma}$



$$\text{Map: } FN := (FN_L, FN_T): T_S \rightarrow (\mathbb{R}_+^*)^g \times \mathbb{R}^g$$

$$\text{length } S_p = 2(1+t)$$

Thm 7.6.3: The map

Thm 7.6.4: Let S be

The function $\ln l_\gamma$:

$$|\ln l_\gamma(\tau_1) - \ln l_\gamma(\tau_2)|$$

pf: $\tau_1, \tau_2 \in T_S, \varphi_i: S \rightarrow$

generates $\langle \gamma \rangle \cong \mathbb{Z}$

$$\pi_\gamma: \tilde{S}_\gamma \rightarrow S_\gamma, (x_i)_\gamma$$

By 3.3.7: $\frac{\pi}{M}$

is quasiconformal, home

$(\varphi_2)_* (\langle \gamma \rangle)$ induces

$$K(f) = K(\tilde{f}_\gamma)$$

ce Thm 7.6.3: The map F_N is a homeomorphism

Thm 7.6.4: Let S be a quasiconformal surface of finite type
The function $\ln h_r: T_S \rightarrow \mathbb{R}$ satisfies:

$$|\ln h_r(T_1) - \ln h_r(T_2)| \leq d(T_1, T_2)$$

Pf: $T_1, T_2 \in T_S$, $\varphi_1: S \rightarrow X_1$, $\varphi_2: S \rightarrow X_2$ (choose s. on γ , γ generates $\langle \gamma \rangle \cong \mathbb{Z} \subseteq \pi_1(S, s)$) (consider covering

$$\pi_r: \tilde{S}_r \rightarrow S; (X_1)_r \rightarrow X_1, (X_2)_r \rightarrow X_2.$$

$$\text{By 3.37: } \frac{\pi}{M} \quad \uparrow \quad M_1 = \frac{\pi}{L_r(T_1)} \quad M_2 = \frac{\pi}{L_r(T_2)} \quad \text{Let } f: X_1 \rightarrow X_2$$

is quasiconformal, homotopic to $\varphi_2^{-1} \circ \varphi_1$, $f(\varphi_1)_* \langle \gamma \rangle =$

$(\varphi_2)_* \langle \gamma \rangle$ induces $\tilde{f}_r: (X_1)_r \rightarrow (X_2)_r$, quasiconformal

$$K(f) = K(\tilde{f}_r)$$

$$\text{By 4.32: } \frac{1}{K} \leq \frac{M}{M'} \leq K$$

$$d(T_1, T_2) := \inf_{f \in \mathcal{H}} \ln K(f) = \inf_r \ln K(\tilde{f}_r) \geq \left| \ln \frac{M_2}{M_1} \right| = |\ln h_r(T_1) - \ln h_r(T_2)|$$

